

Weak forms and problem setup — Nonlinear heat conduction in a two-material 1-D rod

1 Governing Equations

1.1 Geometry and Domains

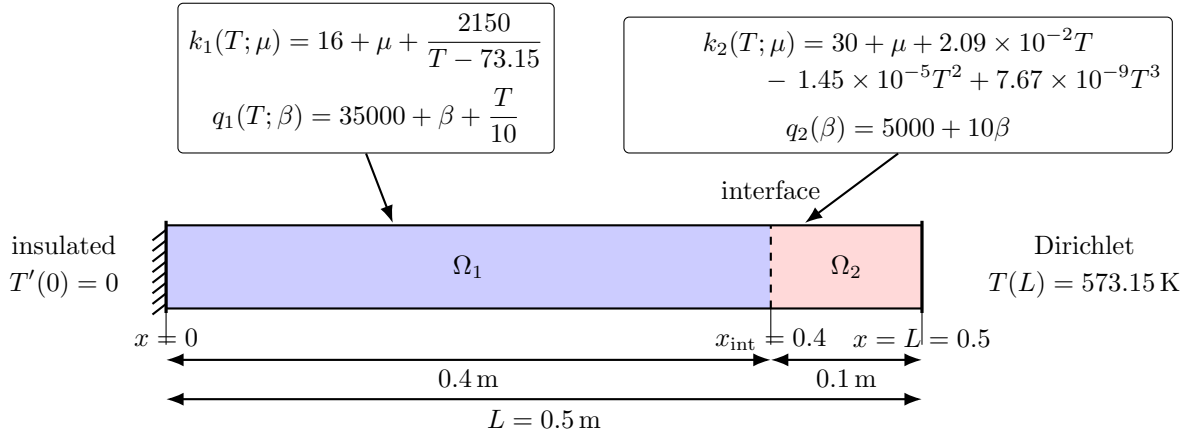


Figure 1: Schematic of the two-region 1-D rod.

We consider steady (time-independent) 1-D conduction through an inhomogeneous rod of length

$$L = 0.5 \text{ m}, \quad \Omega = (0, L).$$

The rod is partitioned at $x_{\text{int}} = 0.4 \text{ m}$ into the sub-domains

$$\Omega_1 = (0, x_{\text{int}}), \quad \Omega_2 = [x_{\text{int}}, L).$$

1.2 Material and Heat Source Parameterization

Both the thermal conductivity k and the volumetric heat source q depend nonlinearly on T :

$$k(x, T; \mu) = \begin{cases} k_1(T; \mu) := 16 + \mu + \frac{2150}{T - 73.15}, & x \in \Omega_1, \\ k_2(T; \mu) := 30 + \mu + 2.09 \times 10^{-2}T - 1.45 \times 10^{-5}T^2 + 7.67 \times 10^{-9}T^3, & x \in \Omega_2, \end{cases} \quad (1)$$

$$q(x, T; \beta) = \begin{cases} q_1(T; \beta) := 35\,000 + \beta + \frac{T}{10}, & x \in \Omega_1, \\ q_2(\beta) := 5\,000 + 10\beta, & x \in \Omega_2. \end{cases} \quad (2)$$

Units: $[k] = \text{W}/(\text{m K})$, $[q] = \text{W}/\text{m}^3$.

1.3 Governing PDE

Define the classical solution space

$$V = \left\{ T \in C([0, L]) \cap C^1([0, L] \setminus \{x_{\text{int}}\}) \mid T|_{\Omega_i} \in C^2(\Omega_i), \ i = 1, 2 \right\}.$$

Find $T \in V$ such that

$$-\frac{d}{dx} \left(k(x, T(x); \mu) T'(x) \right) = q(x, T(x); \beta) \quad \text{in } \Omega, \quad (3)$$

$$T'(0) = 0 \quad (\text{Neumann}), \quad (4)$$

$$T(L) = T_D = 573.15 \text{ K} \quad (\text{Dirichlet}). \quad (5)$$

The dependence of k and q on T makes the problem nonlinear.

2 Weak Form

2.1 Derivation

Let v be any test function with $v(L) = 0$. Multiplying (3) by v , integrating over each sub-domain, and applying integration by parts gives

$$\int_0^L k(x, T; \mu) T' v' dx = \int_0^L q(x, T; \beta) v dx.$$

2.2 Function Spaces

Introduce the Sobolev space

$$H^1(\Omega) = \{ u \in L^2(\Omega) \mid u' \in L^2(\Omega) \}, \quad \|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|u'\|_{L^2}^2.$$

Set

$$V := \{ w \in H^1(\Omega) \mid w(L) = T_D \}, \quad V_0 := \{ v \in H^1(\Omega) \mid v(L) = 0 \}.$$

2.3 Abstract Weak Problem

Given parameters $\boldsymbol{\mu} = (\mu, \beta)$, find $T \in V$ such that

$$a(T, v; \boldsymbol{\mu}) = \ell(T, v; \boldsymbol{\mu}) \quad \forall v \in V_0, \quad (6)$$

with nonlinear forms

$$a(T, v; \boldsymbol{\mu}) := \int_{\Omega} k(x, T; \mu) T' v' dx, \quad (7)$$

$$\ell(T, v; \boldsymbol{\mu}) := \int_{\Omega} q(x, T; \beta) v dx. \quad (8)$$

3 Discretized Forms

3.1 Finite-Element Spaces

Choose conforming subspaces $V_h \subset V$ and $V_{0,h} \subset V_0$ with nodal basis $\{\phi_i\}_{i=1}^N$. The discrete temperature reads

$$T_h(x) = \sum_{i=1}^N T_i \phi_i(x), \quad T_i = T_D \text{ on Dirichlet nodes.}$$

3.2 Nonlinear Stiffness Matrix and Load Vector

$$K_{ij}(T_h; \mu) := \int_{\Omega} k(x, T_h; \mu) \phi'_i \phi'_j dx, \quad (9)$$

$$f_i(T_h; \beta) := \int_{\Omega} q(x, T_h; \beta) \phi_i dx. \quad (10)$$

3.3 Discrete Nonlinear System

The nodal residual becomes

$$R_i(T; \mu, \beta) = \sum_{j=1}^N K_{ij}(T_h; \mu) T_j - f_i(T_h; \beta), \quad i = 1, \dots, N,$$

so the global system is $\mathbf{R}(T; \mu, \beta) = \mathbf{0}$.

Collecting all R_i entries gives the nonlinear vector equation

$$\mathbf{R}(T; \mu, \beta) = \mathbf{0}, \quad \mathbf{R}(T; \mu, \beta) := \mathbf{K}(T; \mu)T - \mathbf{f}(T; \beta).$$

At each Newton (or Picard) iteration we solve the linearized system

$$\mathbf{J}(T^{(m)}; \mu, \beta) \Delta T^{(m)} = -\mathbf{R}(T^{(m)}; \mu, \beta),$$

where $\mathbf{J}(T; \mu, \beta)$ is the Jacobian matrix with entries J_{ij} defined in (11).

4 Residual and Jacobian

4.1 Continuous Residual Functional

For any $v \in V_0$,

$$R(T; \mu, \beta)[v] = \int_{\Omega} k(x, T; \mu) T' v' dx - \int_{\Omega} q(x, T; \beta) v dx.$$

4.2 Jacobian (Fréchet Derivative)

For a perturbation $\delta T \in V_0$,

$$J(T; \mu, \beta)[\delta T, v] = \int_{\Omega} \left(k \delta T' v' + \frac{\partial k}{\partial T} \delta T T' v' - \frac{\partial q}{\partial T} \delta T v \right) dx. \quad (11)$$

4.3 Finite-Element Jacobian Matrix

With basis $\{\varphi_i\}$,

$$J_{ij} = \int_{\Omega} \left(k \varphi'_j \varphi'_i + \frac{\partial k}{\partial T} T'_h \varphi_j \varphi'_i - \frac{\partial q}{\partial T} \varphi_j \varphi_i \right) dx.$$

Closed-form derivatives:

$$\frac{\partial k_1}{\partial T} = -\frac{2150}{(T - 73.15)^2}, \quad \frac{\partial k_2}{\partial T} = 2.09 \times 10^{-2} - 2(1.45 \times 10^{-5})T + 3(7.67 \times 10^{-9})T^2,$$

$$\frac{\partial q_1}{\partial T} = 0.1, \quad \frac{\partial q_2}{\partial T} = 0.$$